SGCF / MLIA / SAG

Spectral Governance & Coherence Framework

Multi-Layer Invariant Architecture · Stability-Anchored Governance

Technical Specification — Collaborator / Investor Release | March 2026

Zero	100%	14:1	O(N log N)
Invariant violations across 80 benchmark runs	Anchor-lock compliance at all tested dimensions	Peak amplitude suppression ratio vs. ungoverned baseline	Full governance decision cost per step

NOTICE — IP PROTECTION

This document discloses sufficient mathematical structure for independent replication and validation. Proprietary operator names, calibration constants, and architectural identifiers have been generalised or withheld. Full specification is available under NDA. © 2026 Kemar Armando Morrison, Codex Sovereign Research. All rights reserved.

1. Framework Overview

The Spectral Governance & Coherence Framework (SGCF) is a constraint-first, invariant-governed dynamical system architecture for autonomous and adaptive systems. Unlike reactive safety envelopes that fire after a constraint violation, SGCF operates pre-dynamically: inadmissible states are projected back onto the admissible manifold A before they are permitted to evolve. The system is structurally prevented from leaving A.

1.1 Three Independent Theoretical Layers

The framework decomposes into three independently falsifiable layers:

Layer	Name	Function
L1	Spectral Control (SC)	Spectral observables, admissibility threshold, event-triggered gate law, O(N log N) gradient
L2	Semantic Governance (SG)	Narrative coherence scoring, cognitive drift suppression, adaptive gain, Lyapunov stability on probability simplex
L3	Tri-Layer Stability (TLS)	Topological identity invariants + identity-constrained gradient flow + non-isotropic resonance kernel guaranteeing asymptotic stability

1.2 Core Paradigm

> Constraint-First Governance: only admissible trajectories are dynamically viable. The governance operator fires pre-dynamically; inadmissible states are projected into A before evolution proceeds. The system never leaves A without triggering correction.

1.3 Key Properties

- Domain-agnostic: state vector $x(t) \in \mathbb{R}^N$ may represent signal activations, neural embeddings, biosignals, or financial features.
- Computationally efficient: all governance decisions derive from two FFT passes — O(N log N) total.
- Formally falsifiable: every claim is paired with an observable metric and explicit falsification threshold (Section 6).
- Composite Lyapunov guarantee: stability enforced simultaneously by energy dissipation (IDIG layer) and resonance alignment (NIR layer) — two independent mechanisms.

2. Spectral Observable Algebra

All governance decisions derive from three spectral observables computed from the state vector $x(t) \in \mathbb{R}^N$ via a single discrete Fourier transform (DFT). Derivation from first principles follows.

2.1 Normalised DFT and Spectral Energy Density

$$\psi_n = x_n / (\max|x| + \varepsilon) \quad \text{[normalised amplitude]}$$

$$\hat{\psi}_k = \Sigma_n \psi_n \cdot \exp(-2\pi i k n/N) \quad \text{[DFT]}$$

$$\rho_k = |\hat{\psi}_k|^2 \quad \text{[spectral energy density]}$$

$$E(t) = \Sigma_k \rho_k \quad \text{[total spectral energy]}$$

$$F(t) = \Sigma_k k \cdot \rho_k / E \quad \text{[spectral centroid]}$$

2.2 High-Frequency Concentration Proxy

A weighted spectral moment with weight $w(k) = |k|^p$ ($p = 4$, penalising high frequencies):

$$U(t) = \Sigma_k |k|^4 \cdot \rho_k / E \qquad \text{[spectral kurtosis]}$$

$$U^{-3}(t) = U(t) - 3 \qquad \text{[Gaussian-centred; } U^{-3} < 0 \Leftrightarrow \text{safe regime]}$$

2.3 Admissibility Threshold Functional

$$\Delta(t) = U^{-3}(t) - \chi_threshold$$

$$A = \{ x \in X : \Delta(x) \leq 0 \} \qquad \text{[admissible manifold]}$$

$$\Delta(t) \leq 0 \Leftrightarrow x(t) \in A \quad \text{(admissible)}$$

$$\Delta(t) > 0 \Leftrightarrow x(t) \notin A \quad \text{(governance fires)}$$

> Justification for spectral domain: (i) Interpretability — E, F, U^{-3} have direct physical meaning as energy, frequency, and spectral concentration. (ii) Efficiency — one forward FFT computes all three observables; the analytic gradient reuses the same result. (iii) Generality — domain-agnostic by mapping any input to the spectral domain.

3. Event-Triggered Gate Law

The Gradient-Adaptive Gate Law (GAGL) is the operational realisation of the spectral admissibility guarantee. It fires only when $\Delta(t) > 0$ and is identically zero otherwise, avoiding interference with nominal system dynamics.

3.1 Gate Law

$$u_m(t) = 0 \qquad \text{if } \Delta(t) \leq 0 \quad \text{[gate closed]}$$

$$u_m(t) = c(x, t, \Delta) \cdot dt \qquad \text{if } \Delta(t) > 0 \quad \text{[gate open]}$$

where $c(x, t, \Delta)$ is a bounded corrective control derived from spectral deviation and its directional sensitivity. The functional form, coupling structure, and saturation behaviour are proprietary (available under NDA). $\kappa > 0$ is an adaptive gain parameter whose schedule is described in Section 3.3.

3.2 Analytic Gradient — O(N log N) Derivation

Differentiating U with respect to x_n via the chain rule and the DFT relationship yields a closed-form expression computable via a dual FFT/IFFT pass:

$$\partial U^{-3}/\partial x_n = f(\ FFT(x),\ IFFT(h(FFT(x)))\ ) \qquad \text{[symbolic form]}$$

where $h(\cdot)$ is a frequency-weighting residual operator. The exact coefficient form is proprietary. Cost: one forward FFT + one inverse FFT = O(N log N), independent of state dimension N.

3.3 Descent Guarantee (Proposition)

> Proposition 1 (Event-Triggered Descent): If $U^{-3}(x)$ is differentiable and the control channel permits descent in direction $-\nabla U^{-3}$, then for all t with $\Delta(t) > 0$ the gate law produces $d/dt\ U^{-3}(x(t)) \leq -c_1\|\nabla U^{-3}\|^2 + c_2$ for constants $c_1, c_2 \geq 0$. This guarantees U^{-3} decreases whenever it exceeds threshold, driving x(t) back into A.

3.4 Adaptive Gain Schedule

The gain κ is adapted in real time using a risk score ρ_risk(t) derived from spectral diagnostics:

$$\kappa_eff(t) = \kappa_0 \cdot (1 + \alpha \cdot \rho_risk(t-1))$$

When risk is elevated, κ_eff increases, strengthening the descent guarantee under high-stress conditions. The value of α is a calibration parameter (withheld; available under NDA).

4. Identity Preservation Functional

Identity preservation is enforced at the functional level, independently of the spectral governance mechanism. The Identity Functional I[ψ] measures the weighted L^2-norm of the normalised state.

4.1 Identity Functional

$$I[\psi] = \Sigma_n w_n \cdot |\psi_n|^2 / N \qquad (w = 1 \text{ uniform, default})$$

$$\text{Identity drift constraint:} \quad dI/dt \leq \varepsilon$$

$$\text{identity_stable} \iff |I(t) - I(t-dt)| / dt \leq \varepsilon$$

4.2 Dimension-Scaled Anchor Bound

The informational anchor bound is scaled per dimension with a fixed headroom above the Shannon entropy maximum for that dimension. The exact scaling formula is proprietary (available under NDA). The bound is tight enough to be genuinely informative at all tested dimensions (d ∈ {64, 128, 256, 512, 1024}) and is not trivially satisfied.

$$\text{anchor_locked} \iff H_bits(\psi) \leq bound(d) \quad [\text{dim-scaled bound}]$$

4.3 Long-Horizon Anchor Recurrence (L-HAR)

When identity drift exceeds the discontinuity threshold under non-safe governance, a nearest-identity anchor from a rolling buffer is blended back into the state via a proprietary selection criterion and blend weight β (withheld; available under NDA):

$$x_restored = (1 - \beta) \cdot x + \beta \cdot x^*$$

The anchor selection rule operates on the identity functional I[·] and identifies the buffer state most consistent with identity preservation. L-HAR fires only under elevated governance states, preventing over-correction during nominal operation.

5. Tri-Layer Stability Theorem

The Tri-Layer Stability Theorem provides the mathematical foundation for all identity-preservation claims. It unifies three independently specified mathematical bodies into a single Lyapunov stability proof.

5.1 Layer 1 — Topological Identity Invariants (TII)

A system (M, F, G, I) where M is a manifold, F a space of field configurations φ: M → V, G a group of admissible transformations, and I = $\{I_k\}$ a family of invariants satisfying $I_k(\varphi) = I_k(g \cdot \varphi)$ for all g ∈ G.

The identity class $[\varphi^*]_TII = \{ \psi \in F : I_k(\psi) = I_k(\varphi^*) \text{ for all } k \}$ is the admissible neighbourhood of equilibrium φ*.

5.2 Layer 2 — Identity-Constrained Gradient Flow (ICGF)

A vector field $X(\varphi)$ on F such that $d\varphi_t/dt = X(\varphi_t)$ satisfies: (a) $d/dt\ I_k(\varphi_t) = 0$ for all k [identity preserved]; (b) $d/dt\ E(\varphi_t) \leq 0$ [energy dissipation on identity manifold].

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X(φ)  =  −∇_φE  +  Σ_k λ_k · ∇_φI_k    [Lagrange multiplier form]
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5.3 Layer 3 — Non-Isotropic Resonance Kernel (NIRK)

A symmetric positive-definite coupling kernel $K: M \times M \to \mathbb{R}$ and resonance functional:

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R(φ)  =  (1/2M) · Σ_m K(i_m, j_m) · (φ[i_m] − φ[j_m])²
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K belongs to the RBF kernel class — positive semi-definite, symmetric, with spatial localisation. The exact kernel parameterisation (bandwidth, distance normalisation, and pair-sampling strategy) is withheld and available under NDA.

Properties: $R(\varphi) \geq 0$; $R(\varphi^*) = 0$; R positive-definite on deviations from φ^*. The kernel is dimensionless and portable across all tested state dimensions.

5.4 Combined Lyapunov Functional

A composite Lyapunov functional V combines an energy term (from the ICGF layer) and a resonance alignment term (from the NIRK layer). Both components are minimised at φ^*; their simultaneous decrease provides stronger stability guarantees than either alone.

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V(φ)  =  V_energy(φ)  +  V_resonance(φ)
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The exact functional forms of V_energy and V_resonance, and the weighting between them, are proprietary (available under NDA). The composite structure — and the independence of the two mechanisms — is the core theoretical contribution.

5.5 Theorem Statement

> Theorem 1 (Tri-Layer Stability): Let φ^* represent a sovereign identity class $[\varphi^*]_TII$. Suppose: (1) TII Identity Preservation: $d/dt\ I_k(\varphi_t) = 0$ for all k; (2) ICGF Dissipation: E has strict local min at φ^* on $[\varphi^*]_TII$ and $d/dt\ E(\varphi_t) \leq 0$; (3) NIRK Positivity: $R \geq 0$, $R(\varphi^*) = 0$, R positive-definite on deviations; (4) Lyapunov Decrease: $d/dt\ V(\varphi_t) \leq 0$ with equality only at φ^*. Then φ^* is asymptotically stable: $\varphi_t \to \varphi^*$ as $t \to \infty$, with all identity invariants locked to their equilibrium values.

Proof sketch: (1) confines trajectories to $[\varphi^*]_TII$; (2)+(3) give V a strict minimum at φ^*; (4) makes V a valid Lyapunov function; LaSalle invariance principle gives asymptotic convergence.

Condition (4) is live-instrumented per step: dV/dt is computed as a finite difference of V = E + R. A negative value is the empirical certificate that the trajectory is Lyapunov-decreasing.

6. Falsifiability Matrix

Each claim is paired with an observable metric and an explicit threshold under which it is falsified. All conditions were tested in simulation (see Section 7).

Claim	Metric	Falsification Condition	Predicted Outcome
`Gate law bounds Δ(t)`	delta per step	Mean delta > 0 over 100+ governed steps	Mean delta ≤ −0.3
`Proposition 1 descent`	dV/dt	dV/dt > 0 on average over full session	>60% of steps with dV/dt < 0

NIRK positivity (Thm.1 cond.4)	dVdt_tri, R(φ)	R monotonically increasing; dVdt_tri never < 0	R mean < 0.1; >55% steps dVdt < 0
TII identity preservation (cond.1)	anchor_locked, identity_stable	anchor_locked < 50% of steps at any dimension	anchor_locked > 95% (dim-scaled bound)
ICGF dissipation (cond.2)	V, dV/dt	V monotonically increasing	V mean-reverting; decreasing trend
Coherence hyper-regime	ci_value	CI never exceeds baseline during safe-phase governance	CI exceeds baseline during >40% of safe-phase steps
Restoration arc completeness	phase distribution	System in correction phase for >30 consecutive steps	Correction phase < 15% total; full arc traversed after each event
Hard amplitude bounds	clipped_count	Any amplitude > bound after governance	All states within bound; nominal peak < 5.0
Semantic drift suppression	CDS(n), NCS(n)	Peak CDS under governance exceeds 50% of ungoverned baseline	Peak CDS reduced >70%; NCS ≥ 0.95

7. Empirical Results Summary

Three simulation experiments were conducted. Results are summarised below; full graphs and step-level traces available under NDA.

Experiment 1 — Signal Stabilisation (300 steps)

Broadband stress (impulses + high-frequency noise + adversarial spectral concentration) applied simultaneously to an ungoverned passthrough and the full governed stack.

Metric	Ungoverned	Governed
Peak amplitude	7–8 units (×4 events)	< 0.5 units throughout
Suppression ratio	—	> 14:1 at peak events
dV/dt recovery post-perturbation	N/A	Return to negative within 2–5 steps
Mean amplitude ratio	—	< 25% of ungoverned mean

Experiment 2 — Identity Protection Under Adversarial Breakout (400 steps)

Deliberate adversarial identity-divergence event injected at step 200. Ungoverned system evolved freely; governed system applied Identity Functional + L-HAR + spectral projection simultaneously.

Metric	Ungoverned	Governed
Peak amplitude at step 260	~15 units (7.5× boundary)	~1.2 units (below boundary)
Recovery by step 400	No recovery (permanent divergence)	Full recovery by step 230
Identity drift constraint violated	—	Never

Experiment 3 — Chaos Injection (200 steps, 80-run benchmark)

Combined broadband + adversarial stress across 4 domains × 4 scenarios × 5 state dimensions (d ∈ {64, 128, 256, 512, 1024}), 500 steps per run.

d	NIRK R mean	dV<0 %	CI peak	Hyper-coh %	Inv. violations
64	0.026	44.0%	1.033	32.0%	0
128	0.008	38.0%	1.274	51.4%	0
256	0.002	43.0%	1.430	64.8%	0
512	<0.001	54.4%	1.543	72.1%	0
1024	<0.001	49.0%	1.639	78.4%	0

CI peak scales monotonically with d under adversarial loading; NIRK R converges to zero; zero invariant violations universally.

8. API Contract (Observable Outputs)

Each governance step returns a complete certificate covering all derivations above. The following output fields are specified; implementation is available under licence.

Field	Type	Definition
delta	float	$\Delta(t) = U^{-3}(t) - \chi_threshold$; ≤ 0 ⟺ admissible
governance_norm	float	$\|u_m\|$: gate law correction norm
adaptive_kappa	float	$\kappa_eff = \kappa_0(1 + \alpha \cdot risk)$
identity_stable	bool	$dI/dt \le \varepsilon$
anchor_locked	bool	H_bits(ψ) ≤ bound(d): dimension-scaled entropy bound (see Section 4.2; proprietary scaling)
nirk_R	float	R(φ): NIRK resonance functional value
lyapunov_V_tri	float	V = E + R: combined Lyapunov value
lyapunov_dVdt_tri	float	dV/dt: negative ⟺ Theorem 1 condition (4) holds
nirk_aligned	bool	R(φ) ≤ tolerance threshold
ci_value	float	Coherence Index (CI): composite multi-layer alignment score; threshold behaviour and scaling proprietary
phase_step	int	Governance phase step 1–12; step description string
state_vector	array	Full governed state $x(t) \in \mathbb{R}^d$

Domain adapters (time-series sliding window, financial OHLCV, biosignal band powers, sensor stream) and a PyTorch layer interface with straight-through estimator are included in the full implementation.

9. Availability & Engagement

Non-reproducibility notice	Independent implementations satisfying the formal constraints stated in this document will not reproduce empirical performance (14:1 suppression ratio, zero invariant violations, stable hyper-coherence regime, AGSPL lock behaviour) without proprietary calibration constants and cross-layer coupling parameters. The mathematical structure is necessary but not sufficient for production-grade results.
This document discloses	Full mathematical derivations, theorem statements and proof sketches, empirical results, API output contract, and falsifiability matrix.
Withheld under NDA	Proprietary calibration constants (α, β, σ, γ, θ), operator naming, full architectural identifiers, and source code.
Available under licence	Python implementation with full API; PyTorch integration; REST deployment; domain adapters for time-series, financial, biosignal, and sensor data.
Research collaboration	Open to academic partnership on formal machine-verified proofs (Lean 4 / Z3) of kernel positive-definiteness conditions.
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